Q#01

```
students=[]
def result(marks):
    total=sum(marks)
    percentage= total/3
    if percentage >=80:
        grade= "A"
    elif percentage>=70:
        grade="B"
    elif percentage>=60:
        grade="C"
    elif percentage>=50;
        grade="D"
    else:
        grade="F"
return total, percentage, grade
for i in range(5):
    name=input("Enter your name:")
    roll =input("Enter roll no:")
    p = int(input("Python marks:"))
    ai = int(input("AI marks:"))
    math=int(input("Math marks:"))

    total, percentage, grade = result([p,ai, math])
    studen= {
      "name": name,
      "roll": roll,
      "total": total,
      "percentage": percentage,
      "grade": grade }
```

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```
udents.append(student)
for s in students:
    print("Name:", s["name"])
    print("Rollno:", s["roll"])
    print("Total:", s["total"])
    print("Percentage:", s["percentage"])
    print("Grade:", s["grade"])

highest = students[0]
for s in students:
    if s["percentage"] > highest["percentage"]:
        highest = s
print("\n Highest Student:", highest["name"])
print("Percentage:", highest["percentage"])
file = open("result.txt", "w")
for s in students:
    file.write(str(s) + "\n")
file.close()
```

Q#02

```
def analysis(sentence):
    words = sentence.split()
    print("Total words:", len(words))
    print("Total characters:", len(sentence))
    longest = words[0]
    shortest = words[0]
    for word in words:
```

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```
    if len(word) > len(longest):
        longest = word
    if len(word) < len(shortest):
        shortest = word
print("Longest word:", longest)
print("Shortest word:", shortest)
vowels = 0
for letter in sentence:
    if letter.lower() in "aeiou":
        vowels = vowels + 1
print("Vowels:", vowels)
unique = set(words)
print("Unique words:" unique)
print("Alphabetical order:" sorted(unique))
file = open("word_analysis.txt", "w")
file.write("Words:" + str(len(words)) + "\n")
file.write("Characters:" + str(len(sentence) + "\n")
file.write("Longest:" + ~~str(len(~~ longest + "\n")
file.write("Shortest:" + shortest + "\n")
file.write("Vowels:" + str(vowels) + "\n")
file.close

sentence = input("Enter a sentence:")
~~analysis~~
analysis(sentence)
```

"grade : grade 5

Q#03:

```
products = {}
def bill():
    total = 0
    for name in products:
        price = products[name][0]
        quantity = products[name][1]
        total = total + price * quantity
    return total

for i in range(5):
    name = input("Product name:")
    price = input("Price:")
    quantity = int(input("Quantity:"))
    products[name] = [price, quantity]

total = bill()
if total > 10000:
    discount = total * 10/100
else:
    discount = 0
final = total - discount
print("Total:", total)
print("Discount:", discount)
print("Final Bill:", final)
file = open("bill.txt", "w")
file.write("Total:" + str(total) + "\n")
file.write("Discount:" + str(discount) + "\n")
file.write("Final bill:" + str(final))
file.close()
```

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Q#04:

```
def add(a,b):
    return a+b
def subtract(a,b):
    return a-b
def multiply(a,b):
    return a*b
def divide(a,b):
    return a/b
def modulus(a,b):
    return a%b

While True:
    a=int(input("Enter the first number:"))
    b=int(input("Enter the second number:"))
    print("+ Addition \n - Subtraction \n * Multiplication \n
          / Division \n % Modulus")
    op=input("Enter operation:")
    if op=="+":
        print(add(a,b))
    elif op=="-":
        print(subtract(a,b))
    elif op=="*":
        print(multiply(a,b))
    elif op=="/"
        print(divide(a,b))
    elif op=="%"
        print(modulus(a,b))
```

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```
else:
    print("Wrong Operation")
again = input("Continue? yes/no:")
if again == "no":
    break.
```

Q#05:

```
menu = {
    "Burger": 500,
    "Pizza": 800,
    "Fries": 250,
    "Drink": 150,}
orders = {}
def bill():
    total=0
    for item in orders:
        total= total + menu[item] *orders[item]
        return total

while True:
    print("1. Burger")
    print("2. Pizza")
    print("3. Fries")
    print("4. Drink")
    print("5. Finish")

    choice = input("Choose item:")
    if choice == "1":
        item= "Burger"
```

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```
elif choice=="2":
    item = "Pizza"
elif choice=="3":
    item="Fries"
elif choice=="4":
    item="Drink"
elif choice=="5":
    break
else:
    print("Wrong Choice")
    continue
quantity=int(input("Enter Quantity:"))
order[items]=quantity
total=bill()
print("\n Final Bill:",total)
file=open("order.txt","w")
file.write("Final Bill:"+str(total))
file.close
```